

SHEPHERD BAY, NU, Canada

WMO: 719110

Lat: 68.817N

Lon: 93.433W

Elev: 43

StdP: 100.81

Time Zone: -7.00 (NAM)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-46.9	-45.2	-45.9	0.0	-40.4	-43.5	0.1	-35.0	16.4	-31.3	14.1	-29.0	3.1	140	1.133

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	18.1	12.9	15.9	11.7	13.8	10.4	12.8	16.5	11.4	14.6	10.1	12.9	5.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.0	8.2	14.0	9.4	7.4	13.0	8.0	6.7	11.8	36.4	16.6	32.8	15.0	29.6	13.0	19.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.2	12.4	11.0	DB	-48.7	21.4	2.4	2.4	-50.4	23.1	-51.8	24.6	-53.2	25.9	-54.9	27.7	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-14.5	-34.4	-35.2	-30.1	-21.1	-9.9	1.3	8.6	6.6	0.2	-9.7	-22.1	-29.7
	DBStd	16.55	6.71	6.34	7.10	6.64	5.28	4.20	3.81	3.61	3.53	5.83	7.18	7.46
	HDD10.0	8990	1378	1265	1243	932	618	263	75	119	294	609	962	1232
	HDD18.3	11987	1636	1498	1501	1182	876	511	303	365	544	868	1212	1490
	CDD10.0	45	0	0	0	0	0	3	31	12	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.2	4.8	4.7	4.9	5.1	5.3	5.0	5.2	5.4	5.8	6.4	5.3	4.8
Precipitation	PrecAvg	155	4	4	5	7	8	16	24	36	25	19	8	5
	PrecMax	242	17	25	31	31	19	60	65	89	64	42	19	13
	PrecMin	81	0	0	0	0	0	0	5	4	2	4	0	0
	PrecStd	42	4	6	7	7	6	15	15	20	15	9	6	4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-12.0	-15.1	-9.4	-3.1	1.0	17.6	21.2	19.2	11.6	1.8	-3.9	-8.2
		MCWB	-12.4	-15.5	-9.8	-3.5	0.4	11.5	14.7	14.1	9.8	1.1	-4.1	-8.4
	2%	DB	-17.7	-18.5	-12.8	-5.6	0.2	13.2	19.0	16.5	8.9	0.3	-6.3	-11.5
		MCWB	-18.0	-18.8	-13.3	-6.3	-0.1	9.5	12.6	13.2	7.2	0.0	-6.8	-11.8
	5%	DB	-20.9	-21.5	-15.5	-8.2	-1.0	10.1	17.2	14.3	6.7	-0.9	-9.0	-14.9
		MCWB	-21.2	-21.9	-15.9	-8.7	-1.4	7.8	12.2	11.2	5.7	-1.2	-9.4	-15.3
	10%	DB	-24.2	-25.2	-18.5	-10.6	-2.5	7.2	15.4	12.4	5.0	-2.1	-11.4	-18.2
		MCWB	-24.5	-25.5	-18.8	-11.0	-3.0	5.6	11.1	10.1	4.2	-2.5	-11.8	-18.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-12.5	-14.8	-12.1	-4.3	0.3	11.6	15.0	15.9	9.7	1.9	-5.5	-10.4
		MCDB	-11.9	-14.4	-11.6	-3.9	0.5	16.8	21.2	19.0	11.3	2.5	-5.2	-10.2
	2%	WB	-17.3	-18.6	-14.9	-7.7	-0.9	10.1	12.9	13.4	6.9	-0.3	-9.4	-12.2
		MCDB	-17.0	-18.3	-14.6	-7.3	-0.5	14.0	17.3	16.4	8.2	-0.1	-9.0	-11.9
	5%	WB	-20.2	-22.2	-17.9	-9.9	-2.8	8.0	11.5	11.8	5.1	-1.8	-12.0	-14.6
		MCDB	-19.8	-22.0	-17.6	-9.6	-2.4	10.5	15.5	14.0	5.8	-1.5	-11.6	-14.2
	10%	WB	-23.9	-25.9	-21.2	-12.2	-4.4	5.7	10.2	10.2	3.4	-3.7	-14.9	-19.7
		MCDB	-23.6	-25.7	-20.9	-11.8	-4.0	7.1	13.4	12.2	4.1	-3.4	-14.6	-19.4
Mean Daily Temperature Range	5% DB	MDBR	7.3	7.1	7.9	8.7	6.6	4.6	7.2	6.0	3.9	5.6	7.1	7.4
		MCDBR	10.2	10.2	9.2	8.4	5.5	8.5	10.4	9.3	7.0	5.2	8.2	8.8
		MCWBR	9.8	9.5	8.4	9.2	6.7	5.5	6.0	6.4	5.7	4.7	7.7	7.9
	5% WB	MCDBR	10.2	9.7	9.4	8.3	6.9	8.3	9.7	8.7	6.4	4.8	8.0	7.8
		MCWBR	10.1	9.4	9.3	8.3	6.8	5.5	6.0	6.0	5.7	5.0	8.1	7.8
Clear-Sky Solar Irradiance	taub	0.149	0.207	0.251	0.304	0.321	0.326	0.351	0.320	0.293	0.233	0.159	N/A	
	taud	1.886	2.054	2.178	2.074	2.121	2.344	2.390	2.509	2.472	2.108	1.890	N/A	
	Ebn at Noon	92	588	788	828	858	871	832	826	751	522	60	0	
	Edh at Noon	5	43	75	115	124	103	96	76	59	40	4	0	
All-Sky Solar Radiation	RadAvg	0.03	0.46	2.00	4.42	6.45	6.00	5.17	3.40	1.68	0.53	0.08	0.00	
	RadStd	0.00	0.03	0.08	0.13	0.27	0.55	0.48	0.30	0.15	0.06	0.01	0.00	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.20	+1.20	N/A	N/A	N/A	N/A	-436	-432	N/A	N/A
Regional (5 neighbors)		+0.71	+0.89	N/A	N/A	N/A	N/A	-301	-292	N/A	N/A

Nomenclature: See separate page